

# Using Databricks Genie for Subreddit Data Analysis

Learn how to utilize the Genie AI assistant in Databricks to generate insights from subreddit data. This guide walks you through querying specific trends, analyzing post distributions, and investigating community activity patterns using natural language.

## 1 Click "Genie"

The screenshot displays the Databricks workspace interface. On the left sidebar, the 'Genie' icon is highlighted with an orange circle. The main workspace area shows a file tree on the left with folders like '.databricks', '02-data-transformation', and '05-iitb-junta-analytics-genie'. The 'deploy' folder is selected, showing files like 'serialized\_space.json' and 'space\_metadata.json'. The central pane shows a code editor with a Python script for deploying a Genie space. The script includes comments and code to get widget values, validate parameters, and print the configuration. The right pane shows the 'Genie Code' interface with a text input field, a 'Run' button, and a 'Genie Code' section displaying the results of the deployment, including the space name, ID, warehouse, and tables.

```
SQL Warehouse ID
warehouse_id
d3754c233dd0e522
```

```
12:09 PM ... 2: Cell 2
```

```
1 # Get widget values
2 warehouse_id = dbutils.widgets.get
  ("warehouse_id")
3
4 # Validate required parameters
5 if not warehouse_id:
6     raise ValueError("warehouse_id is
  required. Please set it in the widget
  above.")
7
8 print(f"Configuration:")
9 print(f" Warehouse ID: {warehouse_id}")
```

```
Configuration:
Warehouse ID: d3754c233dd0e522
```

```
12:12 PM ... 4: Cell 4
```

```
1 import json
2 from pathlib import Path
3
4 # Get the directory where this notebook
  lives
```

Genie Code

Always allow in current thread

Thoughts: Excellent! The Genie space w...

Read results for 1 cell

Thoughts: The cell 6 doesn't have result...

Perfect! 🎉 The IITB Junta Analytics Genie space has been successfully deployed!

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- Warehouse: d3754c233dd0e522 (Serverless Starter Warehouse)
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You can now ask natural language questions about the r/iitbombay subreddit data, explore posting trends, discover popular topics, and analyze community engagement patterns.

[Open Genie Space](#)

1 asset

deploy +46 -22

@ for objects, / for commands, ↑↓ for hist...

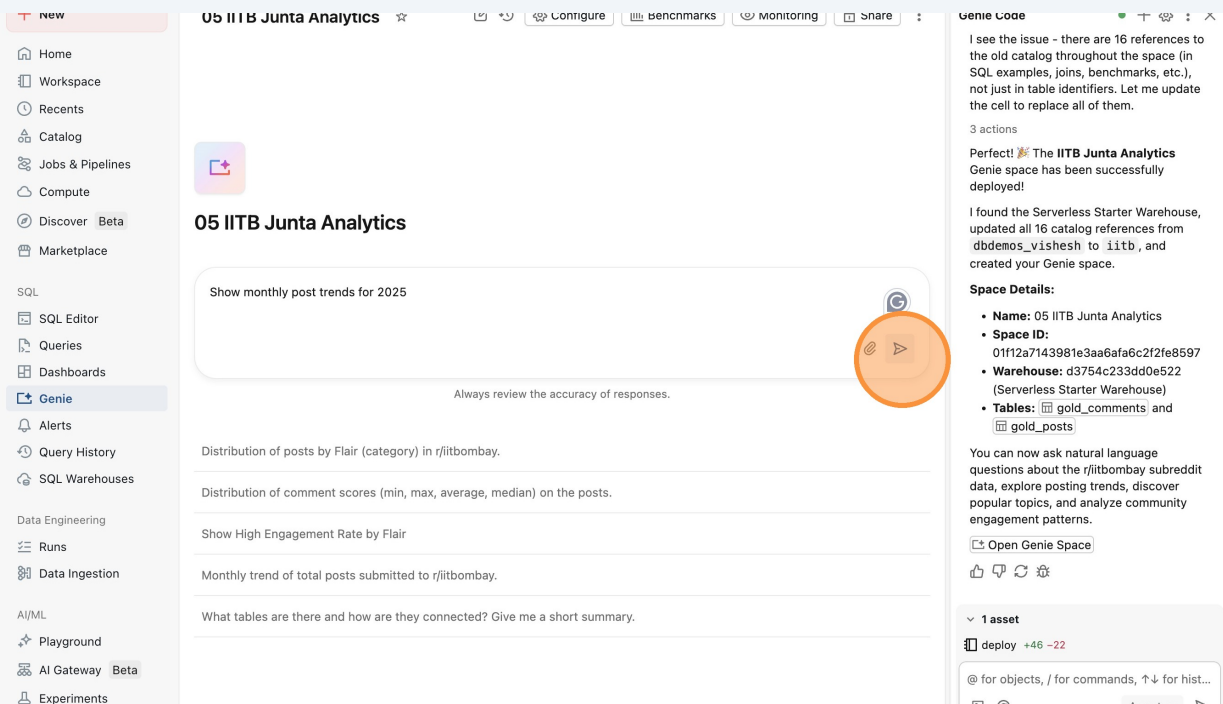
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## 3 Click "Show monthly post trends for 2025"

The screenshot shows the Databricks workspace interface. On the left is a sidebar with navigation options like Home, Workspace, Recents, Catalog, Jobs & Pipelines, Compute, Discover (Beta), Marketplace, SQL, SQL Editor, Queries, Dashboards, Genie (highlighted), Alerts, Query History, SQL Warehouses, Data Engineering, Runs, Data Ingestion, AI/ML, and Playground. The main area displays the '05 IITB Junta Analytics' space. At the top, there's a search bar and a navigation bar with icons for 'Configure', 'Benchmarks', 'Monitoring', and 'Share'. Below the navigation bar, there's a section for 'About this space' with details like Name, Owner, Space ID, and Warehouse. A 'Description' section follows, and then a 'Common questions' section. In the main area, there's a button labeled 'Show monthly post trends for 2025' with an orange circle highlighting it. Below this button, there's a section for '1 asset' with a 'deploy' button showing '+46 -22'. On the right, the 'Genie Code' panel is open, showing a message about updating references to the old catalog. The message says: 'I see the issue - there are 16 references to the old catalog throughout the space (in SQL examples, joins, benchmarks, etc.), not just in table identifiers. Let me update the cell to replace all of them.' Below this, it says 'Perfect! 🎉 The IITB Junta Analytics Genie space has been successfully deployed!' and 'I found the Serverless Starter Warehouse, updated all 16 catalog references from dbdemos\_vishesh to iitb, and created your Genie space.' The 'Space Details' section lists Name, Space ID, Warehouse, and Tables. The 'Tables' section lists 'gold\_comments' and 'gold\_posts'. At the bottom, there's a section for '1 asset' with a 'deploy' button showing '+46 -22'.

## 4 Click this button.



**05 IITB Junta Analytics**

Show monthly post trends for 2025

Always review the accuracy of responses.

Distribution of posts by Flair (category) in r/itbmbay.

Distribution of comment scores (min, max, average, median) on the posts.

Show High Engagement Rate by Flair

Monthly trend of total posts submitted to r/itbmbay.

What tables are there and how are they connected? Give me a short summary.

**Genie Code**

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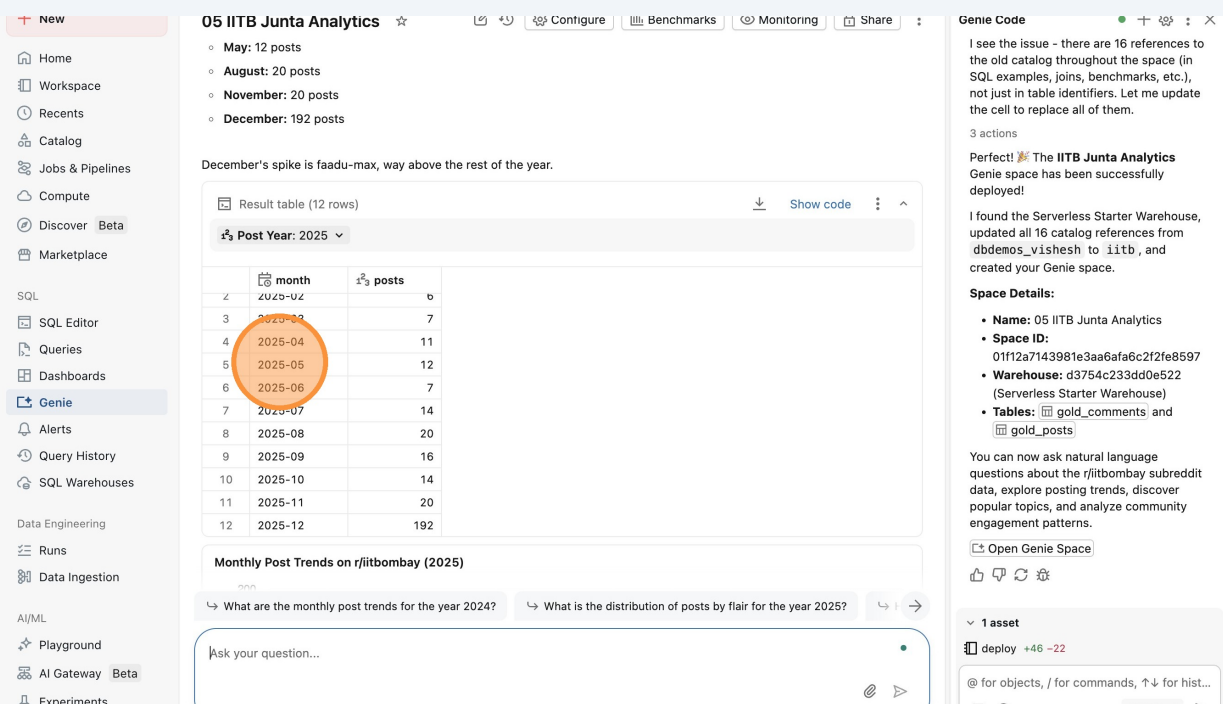
👍 🔄 🗑️

▼ 1 asset

📦 deploy +46 -22

@ for objects, / for commands, ↑↓ for hist...

## 5 Click "2025-05"



**05 IITB Junta Analytics**

- May: 12 posts
- August: 20 posts
- November: 20 posts
- December: 192 posts

December's spike is faadu-max, way above the rest of the year.

Result table (12 rows)

Post Year: 2025

| month   | posts |
|---------|-------|
| 2025-01 | 6     |
| 2025-02 | 7     |
| 2025-04 | 11    |
| 2025-05 | 12    |
| 2025-06 | 7     |
| 2025-07 | 14    |
| 2025-08 | 20    |
| 2025-09 | 16    |
| 2025-10 | 14    |
| 2025-11 | 20    |
| 2025-12 | 192   |

**Monthly Post Trends on r/itbmbay (2025)**

What are the monthly post trends for the year 2024? What is the distribution of posts by flair for the year 2025?

Ask your question...

**Genie Code**

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## 6 Click here.

Workspace

Recents

Catalog

Jobs & Pipelines

Compute

Discover Beta

Marketplace

SQL

SQL Editor

Queries

Dashboards

Genie

Alerts

Query History

SQL Warehouses

Data Engineering

Runs

Data Ingestion

AI/ML

Playground

AI Gateway Beta

Experiments

Features

Got it

|    |         |     |
|----|---------|-----|
|    | 20      |     |
| 10 | 2025-10 | 14  |
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Monthly Post Trends on r/itbombay (2025)

Is this useful? 👍 👎 💬 Share

What are the monthly post trends for the year 2024?

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📄 @ Agent ▾ ▶

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## 7 Click "What is the distribution of posts by flair for the year 2025?"

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Experiments

Features

8

2025-08

20

9

2025-09

16

10

2025-10

14

11

2025-11

20

12

2025-12

192

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| Flair               | Count |
|---------------------|-------|
| 5 Tech              | 6     |
| 6 Survey            | 4     |
| 7 IIT Selection     | 4     |
| 8 Ask me anythin... | 2     |
| 9 Acads             | 2     |
| 10 Polt             | 1     |
| 11 Sports           | 1     |

The bar chart shows the distribution of posts by flair for the year 2025. The x-axis lists flairs: Untagged, Question, Other, Cult, Tech, Survey, IIT Selection, Ask me anythin..., Acads, Polt, and Sports. The y-axis represents the number of posts, ranging from 0 to 200. The 'Untagged' category has the highest count, exceeding 200. An orange circle highlights the 'Ask me anythin...' category in the chart.

Below the chart, there is a search bar with the text 'What are the top flairs by total posts across all years?' and 'What is the distribution of posts by flair for the year 2024?'. The right sidebar shows the 'Genie Code' section with a message: 'Perfect! 🎉 The IITB Junta Analytics Genie space has been successfully deployed!' and 'Space Details' including Name, Space ID, Warehouse, and Tables.

## 9 Click the "Ask your question..." field.

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10

Type "find trends for posting activity by day of week and semester cycles and what can be driving it"

11

Click here.

New

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05 IITB Junta Analytics

Configure
Benchmarks
Monitoring
Share

|    |                 |   |    |
|----|-----------------|---|----|
| 12 | Spring Semester | 5 | 88 |
| 13 | Spring Semester | 6 | 83 |
| 14 | Spring Semester | 7 | 77 |
| 15 | Summer Break    | 1 | 2  |
| 16 | Summer Break    | 2 | 6  |
| 17 | Summer Break    | 3 | 6  |
| 18 | Summer Break    | 4 | 4  |
| 19 | Summer Break    | 5 | 10 |
| 20 | Summer Break    | 6 | 4  |
| 21 | Summer Break    | 7 | 7  |

Posting Activity by Day of Week Across Academic Terms (IITB Reddit)

| Day of Week   | Autumn Semester | Spring Semester | Summer Break |
|---------------|-----------------|-----------------|--------------|
| 1 (Sunday)    | ~40             | ~85             | ~5           |
| 2 (Monday)    | ~45             | ~80             | ~5           |
| 3 (Tuesday)   | ~50             | ~95             | ~5           |
| 4 (Wednesday) | ~55             | ~90             | ~5           |
| 5 (Thursday)  | ~40             | ~85             | ~10          |
| 6 (Friday)    | ~40             | ~80             | ~5           |
| 7 (Saturday)  | ~40             | ~75             | ~10          |

Would you prefer to see posting trends by weekday name instead of numeric values, or by first three months of the month instead?

Ask your question...

Genie Code

+
-

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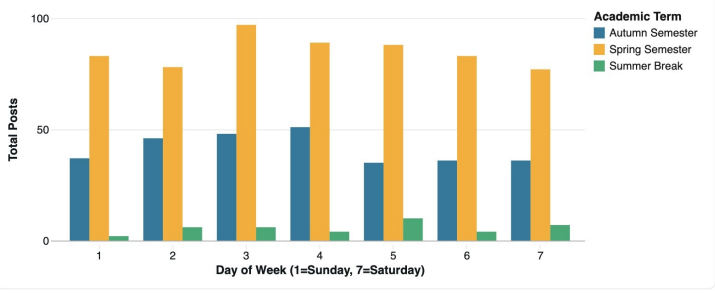


## 12 Click here.

**05 IITB Junta Analytics** ☆      

|    | Academic Term   | Day of Week | Posts |
|----|-----------------|-------------|-------|
| 2  | Autumn Semester | 2           | 46    |
| 3  | Autumn Semester | 3           | 48    |
| 4  | Autumn Semester | 4           | 51    |
| 5  | Autumn Semester | 5           | 35    |
| 6  | Autumn Semester | 6           | 36    |
| 7  | Autumn Semester | 7           | 36    |
| 8  | Spring Semester | 1           | 83    |
| 9  | Spring Semester | 2           | 78    |
| 10 | Spring Semester | 3           | 97    |
| 11 | Spring Semester | 4           | 89    |

**Posting Activity by Day of Week Across Academic Terms (IITB Reddit)**



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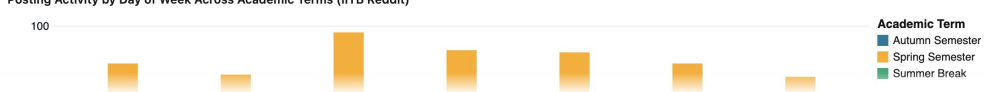
Analysis (click to view) `iitb_subreddit_metrics`

Posting activity is godmax during the Spring Semester, with weekdays 3 and 4 (Wednesday and Thursday) seeing the most posts (97 and 89 respectively). Autumn Semester is ok types, peaking midweek with 51 posts on weekday 4, while Summer Break is total lukka with barely any posts (max 10 on weekday 5). Looks like junta gets enthru to post when acads are in full swing, probably mugging or fighting for fundaes, and chills out during breaks.

Result table (21 rows)

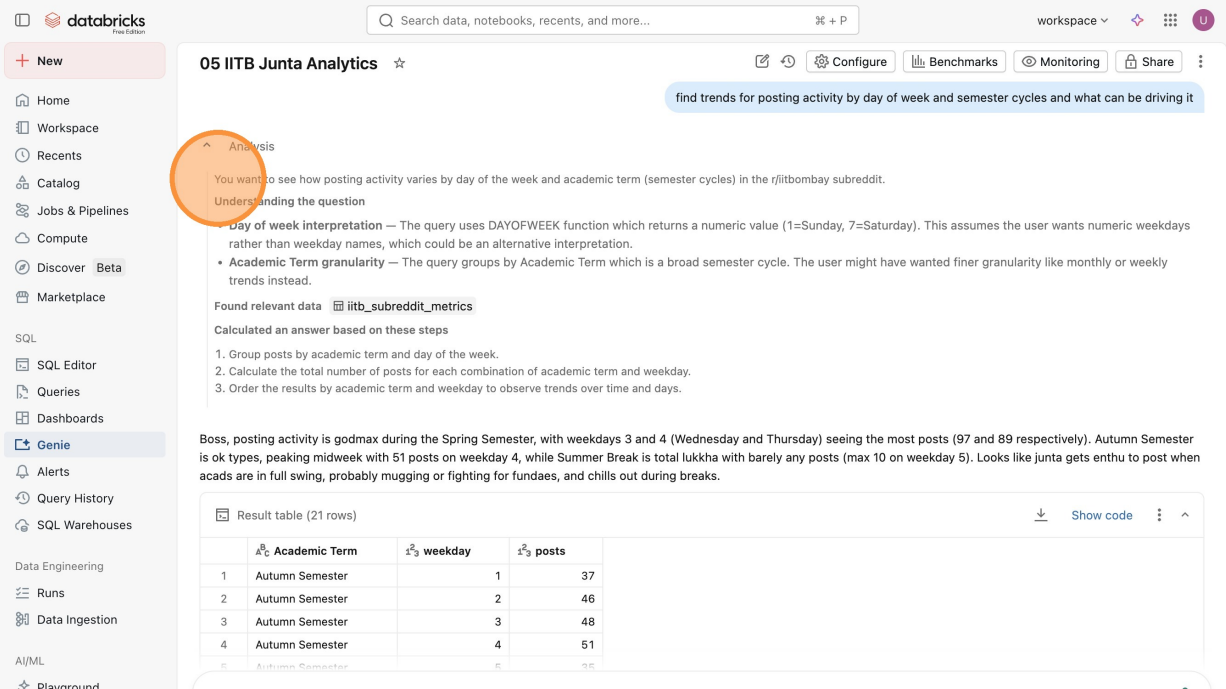
|    | Academic Term   | weekday | posts |
|----|-----------------|---------|-------|
| 1  | Autumn Semester | 1       | 37    |
| 2  | Autumn Semester | 2       | 46    |
| 3  | Autumn Semester | 3       | 48    |
| 4  | Autumn Semester | 4       | 51    |
| 5  | Autumn Semester | 5       | 35    |
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**Posting Activity by Day of Week Across Academic Terms (IITB Reddit)**



14

Click "You want to see how posting activity varies by day of the week and academic term (semester cycles) in the r/iitbombay subreddit."



**05 IITB Junta Analytics** ☆

find trends for posting activity by day of week and semester cycles and what can be driving it

**Analysis**

You want to see how posting activity varies by day of the week and academic term (semester cycles) in the r/iitbombay subreddit.

**Understanding the question**

- Day of week interpretation** — The query uses DAYOFWEEK function which returns a numeric value (1=Sunday, 7=Saturday). This assumes the user wants numeric weekdays rather than weekday names, which could be an alternative interpretation.
- Academic Term granularity** — The query groups by Academic Term which is a broad semester cycle. The user might have wanted finer granularity like monthly or weekly trends instead.

Found relevant data `iitb_subreddit_metrics`

Calculated an answer based on these steps

1. Group posts by academic term and day of the week.
2. Calculate the total number of posts for each combination of academic term and weekday.
3. Order the results by academic term and weekday to observe trends over time and days.

Boss, posting activity is godmax during the Spring Semester, with weekdays 3 and 4 (Wednesday and Thursday) seeing the most posts (97 and 89 respectively). Autumn Semester is ok types, peaking midweek with 51 posts on weekday 4, while Summer Break is total lukkha with barely any posts (max 10 on weekday 5). Looks like junta gets enthu to post when acads are in full swing, probably mugging or fighting for fundaes, and chills out during breaks.

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